MATH4511 Quantitative Methods for Fixed Income Securities
Suggested Solution of Problem Set 1

Problem 1

(a) Using the yield rate, the current price of the bond is given by

$$P_0 = \sum_{k=1}^{24} \frac{1000\left(\frac{0.06}{4}\right)}{(1.05)^{\frac{k}{4}}} + \frac{1000}{(1.05)^6} \approx 1056.409.$$

(*Note: The coupons are paid at times $\frac{1}{4}, \frac{2}{4}, \ldots, \frac{24}{4}$.)

(b) Note that 10 coupons have been paid in the first 30 months and there are 14 unpaid coupons. Also the annual effective yield rate becomes 4.6%. So the dirty price of the bond is

$$P_{30/12}^{dirty} = \sum_{k=11}^{24} \frac{1000\left(\frac{0.06}{4}\right)}{(1.046)^{\frac{k}{4}-\frac{30}{12}}} + \frac{1000}{(1.046)^{6-\frac{30}{12}}} \approx 1047.573.$$

Since the current time is *10th coupon payment* and there is no accrued interest. Hence, the clean price equals dirty price. That is, $P_{30/12}^{clean} = P_{30/12}^{dirty}$.

Problem 2

(a) We first determine the value of unknown n. Note that

- The annual nominal yield rate after 15 months remains to be 6% (compounded annually). This implies that the annual effective yield rate of the bond is also 6%.
- The amount of accrued interest is $100(0.05)\left(\frac{3}{12}\right) = 1.25$ (3 months has passed since the last coupon payment)

Combining these facts, we deduce the following equation:

$$\underbrace{\sum_{k=2}^{n} \frac{100(0.05)}{(1.06)^{k-\frac{15}{12}}} + \frac{100}{(1.06)^{n-\frac{15}{12}}}}_{bond\ price\ (dirty\ price)} - \underbrace{1.25}_{\substack{accrued \\ interest}} = \underbrace{95.94321}_{clean\ value} \Rightarrow \cdots \Rightarrow n = 6.$$

Therefore, the bond price after 3 years (3 coupons have been paid) is

$$P_3 = \sum_{k=4}^{n} \frac{100(0.05)}{(1.06)^{k-3}} + \frac{100}{(1.06)^{n-3}} = 97.32699.$$

(b) Note that the amount of accrued interest is $100(0.05)\left(\frac{1}{12}\right) = 0.416667$ (1 month has passed since the last coupon payment). Thus, the clean price of the bond at that time is

$$P_{\frac{25}{12}}^{Clean} = \underbrace{\sum_{k=3}^{n} \frac{100(0.05)}{(1.06)^{k-\frac{25}{12}}} + \frac{100}{(1.06)^{n-\frac{25}{12}}}}_{bond\ price\ (dirty\ price)} - \underbrace{0.416667}_{\substack{accrued \\ interest}} \approx 96.58812.$$

Problem 3

Firstly, the current price of the bond (at time 0) is

$$P_0 = \sum_{k=1}^{16} \frac{800\left(\frac{0.04}{2}\right)}{(1.05)^{\frac{k}{2}}} + \frac{800}{(1.05)^8} = 750.8481.$$

We let i' be the annual effective interest rate after 3 years. Suppose that $i' = 5\%$, the bond price at that time should be

$$P_3|_{i'=5\%} = \sum_{k=7}^{16} \frac{800\left(\frac{0.04}{2}\right)}{(1.05)^{\frac{k}{2}-3}} + \frac{800}{(1.05)^{8-3}} = 767.0749.$$

However, it is given that $P_3 < P_0$. It follows that $P_3 < P_0 < P_3|_{i'=5\%}$. Together with the fact the bond price is decreasing with respect to yield rate, we get $i' > 5\%$. The answer is **C**.

Problem 4

(a) The current price of the bond is

$$P_0 = \frac{1000(0.06)}{(1+0.02)^1} + \frac{1000(0.06)+1000}{(1+0.023)^2} \approx 1071.696.$$

We let i be the annual effective yield rate of the bond, Then i satisfies

$$\frac{1000(0.06)}{(1+i)^1} + \frac{1000(0.06)+1000}{(1+i)^2} = P_0 \approx 1071.696.$$

Solving the equation (via. quadratic formula), we get

$$i \approx 0.022915.$$

(b) To facilitate the analysis, we let

- $i^{(\infty)}(t)$ be the t-year spot rate which is quoted as annual nominal interest rate compounded continuously.
- $i^{(2)}(t)$ be the t-year spot rate which is quoted as annual nominal interest rate compounded semi-annually.

Recall that $s_0(t)$ quoted in the table is annual effective interest rate, one can deduce that

$$s_0(t) = e^{i^{(\infty)}(t)} - 1 \Rightarrow i^{(\infty)}(t) = \ln\left(1 + s_0(t)\right) \quad and$$

$$s_0(t) = \left(1 + \frac{i^{(2)}(t)}{2}\right)^2 - 1 \Rightarrow i^{(2)}(t) = 2\left(\left(1 + s_0(t)\right)^{\frac{1}{2}} - 1\right).$$

Hence, the corresponding set of spot rates can be computed as

	3 months	6 months	12 months	18 months	24 months
$s_0(t)$	1.5%	1.6%	2%	2.2%	2.3%
$i^{(\infty)}(t)$	1.4889%	1.5873%	1.9803%	2.1761%	2.2739%
$i^{(2)}(t)$	1.4944%	1.5937%	1.9901%	2.188%	2.2869%

Problem 5

Using the spot rate quoted at time 0 (in annual effective interest rate), then the current bond price is found

$$P_0 = \frac{500\left(\frac{0.036}{2}\right)}{(1.016)^{0.5}} + \frac{500\left(\frac{0.036}{2}\right)}{(1.02)^{1}} + \frac{500\left(\frac{0.036}{2}\right)}{(1.022)^{1.5}} + \frac{500 + 500\left(\frac{0.036}{2}\right)}{(1.023)^{2}} \approx 512.8331.$$

After 1 years, the spot rates (increased by 0.2%) become

	3 months	6 months	12 months	18 months	24 months
$s_0(t)$	~~1.5%~~ 1.7%	~~1.6%~~ 1.8%	~~2%~~ 2.2%	~~2.2%~~ 2.4%	~~2.3%~~ 2.5%

Then the price of on-going bond can be computed as

$$P_1 = \frac{500\left(\frac{0.036}{2}\right)}{(1.018)^{0.5}} + \frac{500 + 500\left(\frac{0.036}{2}\right)}{(1.022)^{1}} \approx 506.9631.$$

Remark of Problem 5

Alternatively, one can calculate the bond price (at time 0) using the corresponding annual nominal interest rate as derived in Problem 4(b). For example:

- If $i^{(\infty)}(t)$ (continuous compounding) is adopted, the bond price can be computed as

$$P_0 = \frac{500\left(\frac{0.036}{2}\right)}{e^{0.015873(0.5)}} + \frac{500\left(\frac{0.036}{2}\right)}{e^{0.019803(1)}} + \frac{500\left(\frac{0.036}{2}\right)}{e^{0.021761(1.5)}} + \frac{500 + 500\left(\frac{0.036}{2}\right)}{e^{0.022739(2)}} \approx 512.8335.$$

- If $i^{(2)}(t)$ (semi-annually compounding) is adopted, the bond price can be computed as

$$P_0 = \frac{500\left(\frac{0.036}{2}\right)}{\left(1 + \frac{0.015937}{2}\right)} + \frac{500\left(\frac{0.036}{2}\right)}{\left(1 + \frac{0.019901}{2}\right)^{2}} + \frac{500\left(\frac{0.036}{2}\right)}{\left(1 + \frac{0.02188}{2}\right)^{3}} + \frac{500 + 500\left(\frac{0.036}{2}\right)}{\left(1 + \frac{0.022869}{2}\right)^{4}} \approx 512.8333.$$

(*Note: The answers are slightly different due to the rounding error)

(*Note: One can perform a similar calculation at time 1 by converting the spot rates in the above table into equivalent annual nominal yield rate.)

Problem 6

(a) We let $s_0(t)$ be t year spot rates quoted as annual effective yield rate. Using the bond pricing formula and the given information, we deduce that

$$\frac{1000}{\left(1 + s_0(1)\right)^{1}} = \underbrace{970.8738}_{P_A},$$

$$\frac{1200(0.03)}{\left(1 + s_0(1)\right)^{1}} + \frac{1200(0.03) + 1200}{\left(1 + s_0(2)\right)^{2}} = \underbrace{1200}_{P_B},$$

$$\frac{1200(0.05)}{\left(1 + s_0(1)\right)^{1}} + \frac{1200(0.05)}{\left(1 + s_0(2)\right)^{2}} + \frac{1200(0.05) + 1200}{\left(1 + s_0(3)\right)^{3}} = \underbrace{1264.5347}_{P_C},$$

$$\frac{1400(0.03)}{\left(1 + s_0(1)\right)^{1}} + \frac{1400(0.03)}{\left(1 + s_0(2)\right)^{2}} + \frac{1400(0.03)}{\left(1 + s_0(3)\right)^{3}} + \frac{1400(0.03) + 1400}{\left(1 + s_0(4)\right)^{4}} = \underbrace{1426.8603}_{P_D}.$$

By solving the equations (from the first one to the last one), we get

$$s_0(1) = 0.03, \qquad s_0(2) = 0.03, \qquad s_0(3) = 0.031, \qquad s_0(4) = 0.024649.$$

(b) We let $s_0(t)$ be the spot rate (quoted as annual effective interest rate). Using the given information, we deduce that

(*Be careful that the yield rate quoted in the table is annual nominal yield rate compounded continuously)

$$\frac{1000}{\big(1+s_0(1)\big)^1} = \underbrace{\frac{1000}{e^{0.046}}}_{P_A} \Rightarrow s_0(1) = 0.047074$$

$$\frac{1000}{\big(1+s_0(3)\big)^3} = \underbrace{\frac{1000}{e^{0.055(3)}}}_{P_C} \Rightarrow s_0(3) = 0.056541$$

$$\frac{1000(0.06)}{\big(1+s_0(1)\big)^1} + \frac{1000(0.06)+1000}{\big(1+s_0(2)\big)^2} = \underbrace{\frac{1000(0.06)}{e^{0.049}} + \frac{1000(0.06)+1000}{e^{0.049(2)}}}_{P_B}$$

$$\Rightarrow s_0(2) = 0.0503142.$$

Problem 7

Firstly, the annual effective yield rate of the bond is

$$i = \left(1 + \frac{0.0249}{2}\right)^2 - 1 \approx 0.025055 > s_0(0.5).$$

Suppose that the term structure is flat/inverted (i.e. $s_0(T_1) \geq s_0(T_2)$ for $T_1 < T_2$). Recall that the spot rates satisfy (using pricing formula of bond)

$$\underbrace{\sum_{k=1}^{20} \frac{rF}{(1+i)^{\frac{k}{2}}} + \frac{C}{(1+i)^{10}}}_{\substack{Bond\ price \\ (by\ yield\ rate)}} = \underbrace{\sum_{k=1}^{20} \frac{rF}{\left(1+s_0\left(\frac{k}{2}\right)\right)^{\frac{k}{2}}} + \frac{C}{\big(1+s_0(10)\big)^{10}}}_{\substack{Bond\ price \\ (by\ spot\ rates)}} \overset{s_0\left(\frac{k}{2}\right)<s_0(0.5)<i}{\lessgtr} \sum_{k=1}^{20} \frac{rF}{(1+i)^{\frac{k}{2}}} + \frac{C}{(1+i)^{10}}.$$

So the above equation does not hold and the term structure must be normal (i.e. $s_0(T_1) \leq s_0(T_2)$ for $T_1 < T_2$.).

Problem 8

(a) We consider the following portfolio:

- ✓ Buy 4 units of the second bond and
- ✓ Short 2 units of the first bond.

Then the coupon received at each year is $4\big(100(0.04)\big) - 2\big(100(0.08)\big) = 0$ and the amount received at the end of 10th year is $4(100) - 2(100) = 200$. So it is equivalent to 10-year zero coupon bond with face value \$200.

Using no arbitrage principle, the current price of the 10-year zero coupon bond equals the current value of the portfolio. That is,

$$Bond\ Price = 4(95) - 2(105) = 170.$$

(b) We consider the following portfolio: Buy x unit of the first bond and y unit of the second bond. To construct the replicating portfolio, we choose x and y such that the future payoff of the portfolio matches the future payoff of the coupon bond. That is,

$$\underbrace{100(0.08)x + 100(0.04)y = 500(0.06)}_{Matching\ coupon\ payments} \Rightarrow 2x + y = 7.5$$

$$\underbrace{[100(0.08) + 100]x + [100(0.04) + 100]y = 500 + 500(0.06)}_{Matching\ last\ payment} \Rightarrow x + y = 5.$$

By solving the equations, we get $x = 2.5$ and $y = 2.5$.

Using no arbitrage pricing principle, the current price of the coupon bond equals the value of the portfolio. That is,

$$Bond\ Price = 2.5(105) + 2.5(95) = 500.$$

Problem 9

(a) The modified duration of the 8-year bond (in annual effective yield rate) is

$$D_{Mod}(i) = \frac{1}{1+i} D_{Mac} = \frac{1}{1+0.07} \underbrace{\left(\frac{\sum_{j=1}^{16} \frac{j}{2} \left(\frac{100\left(\frac{0.08}{2}\right)}{(1.07)^{\frac{j}{2}}} \right) + 8\frac{100}{(1.07)^8}}{\sum_{j=1}^{16} \frac{100\left(\frac{0.08}{2}\right)}{(1.07)^{\frac{j}{2}}} + \frac{100}{(1.07)^8}} \right)}_{=6.130951} \approx 5.72986.$$

(b) We let i be the annual effective interest rate of the 10-year bond. Then i must satisfy

$$\sum_{j=1}^{10} \frac{100(0.06)}{(1+i)^j} + \frac{100}{(1+i)^{10}} = 92.$$

By solving the equation numerically, we get $i = 0.071468$.

Using similar method as in (a), the modified duration can be computed as

$$D_{Mod}(i) = \frac{1}{1+i} D_{Mac} = \frac{1}{1+0.071468} \underbrace{\left(\frac{\sum_{j=1}^{10} j\left(\frac{100(0.06)}{(1.071468)^j} \right) + 10\frac{100}{(1.071468)^{10}}}{\sum_{j=1}^{10} \frac{100(0.06)}{(1.071468)^j} + \frac{100}{(1.071468)^{10}}} \right)}_{=7.695597}$$

$$\approx 7.182296$$

Problem 10

We first compute the modified duration and modified convexity with respect to annual nominal interest rate. Note that

$$D_{Mod}(i_0) = \frac{1}{1+\frac{i_0}{2}} D_{Mac} = \frac{1}{1+0.018} \underbrace{\left(\frac{\sum_{j=1}^{24} \frac{j}{2} \left(\frac{1200\left(\frac{0.04}{2}\right)}{(1.018)^j} \right) + 12\frac{1200}{(1.018)^{24}}}{\sum_{j=1}^{24} \frac{1200\left(\frac{0.04}{2}\right)}{(1.018)^j} + \frac{1200}{(1.018)^{24}}} \right)}_{=9.698951} \approx 9.527457.$$

$$C_{Mod}(i_0) = \frac{\sum_{j=1}^{24} \frac{j}{2}\left(\frac{j+1}{2}\right)\left(\frac{1200\left(\frac{0.04}{2}\right)}{(1.018)^j}\right) + \frac{24}{2}\left(\frac{25}{2}\right)\frac{1200}{(1.018)^{24}}}{(1+0.018)^2\left(\sum_{j=1}^{24}\frac{1200\left(\frac{0.04}{2}\right)}{(1.018)^j} + \frac{1200}{(1.018)^{24}}\right)} \approx 108.3235$$

On the other hand, the current bond price is given by

$$P(i_0) = \sum_{j=1}^{24}\frac{1200\left(\frac{0.04}{2}\right)}{(1.018)^j} + \frac{1200}{(1.018)^{24}} = 1246.439.$$

Using second order approximation, the percentage change in the bond price is estimated to be

$$\frac{P(i_0+\varepsilon)-P(i_0)}{P(i_0)} \approx -D_{Mod}\varepsilon + \frac{C_{Mod}}{2}\varepsilon^2 \overset{\varepsilon=0.008}{\approx} -0.06929.$$

Thus, the new bond price can be estimated as

$$P(i_0+\varepsilon) = (1-0.06929)P(i_0) \approx 1160.077.$$

(*Note: The actual bond price is 1155.618.)

Problem 11

Since the coupon is paid annually, the annual effective yield rate equals the annual nominal yield rate of the bond. So when the annual nominal yield rate is increased by 0.15%, the annual effective yield rate is also increased by 0.15%.

We let D_{Mod} be the modified duration of the bond. According to the given information, D_{Mod} satisfies

$$\frac{P(i_0+\varepsilon)-P(i_0)}{P(i_0)} \approx -D_{Mod}(-0.002) = 0.015086 \Rightarrow D_{Mod} = 7.543.$$

We let c be the annual coupon rate. Then we have

$$\underbrace{\frac{1}{1+0.084}\left(\frac{\sum_{i=1}^{12} i\left(\frac{4800c}{(1.084)^i}\right) + 12\left(\frac{4800}{(1.084)^{12}}\right)}{\sum_{i=1}^{12}\frac{4800c}{(1.084)^i} + \frac{4800}{(1.084)^{12}}}\right)}_{D_{Mod}=\frac{1}{1+i}D_{Mac}} = 7.543 \Rightarrow c \approx 0.075$$

So the modified convexity can be found to be

$$C_{Mod} = \frac{\sum_{i=1}^{12} i(i+1)\left(\frac{4800c}{(1.084)^i}\right) + 12(13)\left(\frac{4800}{(1.084)^{12}}\right)}{(1.084)^2\left(\sum_{i=1}^{12}\frac{4800c}{(1.084)^i} + \frac{4800}{(1.084)^{12}}\right)} \overset{c=0.075}{=} 78.13145.$$

By applying second order approximation, the percentage change is estimated to be

$$\frac{P(i_0+\varepsilon)-P(i_0)}{P(i_0)} \approx -D_{Mod}\varepsilon + \frac{C_{Mod}}{2}\varepsilon^2 \overset{\varepsilon=0.0015}{\approx} -0.01114.$$

Problem 12

(a) Since the annual coupon rate for 1st coupon payment is $R_{\left[0,\frac{1}{2}\right]} + 0.15\%$. Since the effective interest rate $R_{\left[0,\frac{1}{2}\right]}$ is $R_{\left[0,\frac{1}{2}\right]} = (1.02)^{\frac{1}{2}} - 1 = 0.00995$. So the amount of first coupon payment is $(0.00995 + 0.0015)100 = 1.14505$.

(b) Recall that the entire floating rate bond can be decomposed into two components:

- Floating rate note with coupon rate $R_{\left[\frac{i-1}{2},\frac{i}{2}\right]}$ and
- An annuity which pay $0.0015(100) = 0.15$ every 6 months for 6 years.

Since the price of the floating rate note in the first component is $F = 100$. Thus, the current price of the floating rate bond is

$$P_0 = 100 + \sum_{k=1}^{12} \frac{0.15}{(1.02)^{\frac{k}{2}}} \approx 101.68879.$$

(c) We shall compute the bond price using the decomposition in (b):

- The value (price) of the annuity (second component) is the present value of all unpaid payments (6 payments remaining). That is,

$$P_{40/12}^{annuity} = \sum_{k=7}^{12} \frac{0.15}{(1.026)^{k/2-40/12}} = 0.86807.$$

- For the first component (FRN), we note that the investor will receive a coupon payment $R_{[3,3.5]}F = \left((1 + 0.025)^{\frac{1}{2}} - 1\right)(100) = 1.2422837$ and an on-going FRN (with value $F = 100$) in the next coupon payment date. Thus, the current value of this component is

$$P_{40/12}^{FRN} = \frac{1.2422837 + 100}{(1.026)^{\frac{42}{12}-\frac{40}{12}}} = 100.8101.$$

Therefore, the current price of the entire floating rate bond is

$$P_{40/12} = P_{40/12}^{annuity} + P_{40/12}^{FRN} = 101.6782.$$

Problem 13

(a) We let $L_{[0,T]}$ be T-year spot rate (quoted as annual effective interest rate). Recall that the fact that $\left(1 + R_{[0,T_1]}\right)\left(1 + K_{[T_1,T_2]}\right) = \left(1 + R_{[0,T_2]}\right)$.

- By taking $T_1 = 0, T_2 = 1$, we get $(1 + 0.026)^1 = \left(1 + L_{[0,1]}\right)^1 \Rightarrow L_{[0,1]} = 0.026$.
- By taking $T_1 = 1, T_2 = 2$, we get
$$(1 + 0.026)^1(1 + 0.028001)^1 = \left(1 + L_{[0,2]}\right)^2 \Rightarrow L_{[0,2]} = 0.027.$$
- By taking $T_1 = 2, T_2 = 3$, we get
$$(1 + 0.027)^2(1 + 0.028501)^1 = \left(1 + L_{[0,3]}\right)^3 \Rightarrow L_{[0,3]} = 0.0275.$$
- By taking $T_1 = 3, T_2 = 4$, we get
$$(1 + 0.0275)^3(1 + 0.033513)^1 = \left(1 + L_{[0,4]}\right)^4 \Rightarrow L_{[0,4]} = 0.029.$$

- By taking $T_1 = 3, T_2 = 4$, we get

$$(1+0.029)^4(1+0.039039)^1 = \left(1+L_{[0,5]}\right)^5 \Rightarrow L_{[0,5]} = 0.031.$$

Remark

You can also express the spot rate in forms of annual nominal interest rate.

(b) The amount of a coupon payment is $cF = 0.04(1000) = 40$.

Using spot rate (obtained in (a)), the current price of the bond is

$$P_0 = \frac{40}{(1+0.026)^1} + \frac{40}{(1+0.027)^2} + \frac{40}{(1+0.0275)^3} + \frac{1040}{(1+0.029)^4} = 1041.408.$$

Using forward rate and the formula in Example 16 in Lecture Note 1, the current price of the bond can be computed as

$$\begin{aligned} P_0 = {} & \frac{40}{(1+0.026)^1} + \frac{40}{(1+0.026)^1(1+0.028001)^1} \\ & + \frac{40}{(1+0.026)^1(1+0.028001)^1(1+0.028501)^1} \\ & + \frac{1040}{(1+0.026)^1(1+0.028001)^1(1+0.028501)^1(1+0.033513)^1} \\ & \approx 1041.408. \end{aligned}$$

Problem 14

(*In this problem, we assume that all interest rates are quoted as annual effective interest rate)

We let $s_0(T)$ be the T-year spot rate (quoted as annual effective interest rate). Then we have $s_0(T_1) < s_0(T_2)$ for any $T_1 < T_2$.

- We let i be the annual effective yield rate of the 5-year coupon bond in (2). Assuming that the bond pays m coupons in a year, then i satisfies

$$\sum_{j=1}^{5m} \frac{rF}{(1+i)^{\frac{j}{m}}} + \frac{F}{(1+i)^5} = \sum_{j=1}^{5m} \frac{rF}{\left(1+s_0\left(\frac{j}{m}\right)\right)^{\frac{j}{m}}} + \frac{F}{\left(1+s_0(5)\right)^5}$$

$$> \sum_{j=1}^{5m} \frac{rF}{\left(1+s_0(5)\right)^{\frac{j}{m}}} + \frac{F}{\left(1+s_0(5)\right)^5}$$

  This implies that $i < s_0(5)$.

- Recall that the forward rate $f_{[4.75,5]}$ (quoted as annual effective interest rate) satisfies

$$\underbrace{\left(1+s_0(4.75)\right)^{4.75}}_{\left(1+R_{[0,4.75]}\right)} \underbrace{\left(1+f_{[4.75,5]}\right)^{0.25}}_{\left(1+K_{[4.75,5]}\right)} = \underbrace{\left(1+s_0(5)\right)^5}_{\left(1+R_{[0,5]}\right)}$$

$$\Rightarrow f_{[4.75,5]} = \left[\frac{\left(1+s_0(5)\right)^5}{\left(1+s_0(4.75)\right)^{4.75}}\right]^{\frac{1}{0.25}} - 1 > \left[\frac{\left(1+s_0(5)\right)^5}{\left(1+s_0(5)\right)^{4.75}}\right]^{\frac{1}{0.25}} - 1 = s_0(5).$$

Hence, we conclude that $f_{[4.75,5]} > s_0(5) > i$.

If the term structure is flat (i.e. $s_0(T) = c$ for all T), we have $f_{[4.75,5]} = s_0(5) = i$,
If the term structure is inverted (i.e. $s_0(T_1) > s_0(T_2)$ for any $T_1 < T_2$.), we have $f_{[4.75,5]} < s_0(5) < i$.

Problem 15

(a) The delivery price of the forward contract is found to be

$$K = \frac{S_0}{B(0;0.5)} = \frac{30}{1/(1.04)^{0.5}} \approx 30.59412.$$

(b) Given the price of the asset after 3 months ($S_{0.25}$), the value of the forward contract is

$$f_{0.25} = S_{0.25} - KB(0.25;0.5).$$

Suppose that the 3-month spot rate is 4% and $S_{0.25} = 35$, we have

$$f_{0.25} = 35 - \frac{30.59412}{(1.02)^{0.25}} = 4.556967.$$

If $S_{0.25} = 25$, the corresponding value of the forward contract is

$$f_{0.25} = 25 - \frac{30.59412}{(1.02)^{0.25}} \approx -5.44303.$$

Problem 16

(a) Recall the fact that $\big(1 + R_{[0,T_1]}\big)\big(1 + K_{[T_1,T_2]}\big) = \big(1 + R_{[0,T_2]}\big)$. By taking $T_1 = 1.5$ and $T_2 = 2$, we get

$$(1 + 0.0475)^{1.5}\big(1 + f_{[1.5,2]}\big)^{0.5} = (1 + 0.05)^2 \Rightarrow f_{[1.5,2]} \approx 0.057536.$$

(b) Recall that the current value of FRA can be expressed as

$$f_0 - NB(0;1.5) = -N\big(1 + K_{[1.5,2]}\big)B(0;2).$$

Since $1 + K_{[1.5,2]} = (1.06)^{\frac{1}{2}}$, we get

$$f_0 = \frac{10000}{(1.0475)^{1.5}} - \frac{10000(1.06^{0.5})}{(1.05)^2} \approx -10.8607.$$

Problem 17

(a) We consider the following portfolio:

Portfolio 1: T-year forward contract in long position and buy T-year zero coupon bond with face value K.

Portfolio 2: One unit of the underlying asset and short t_D-year zero coupon bond with face value D (The second strategy will generate a negative cashflow D at time t_D)

One can check that both portfolios will generate one unit of asset at time T.

By no arbitrage pricing principle, the current value of the two portfolios must be the same. That is,

$$f_0 + KB(0;T) = S_0 - DB(0;t_D).$$

By setting $f_0 = 0$, we deduce that

$$K = \frac{S_0 - DB(0;t_D)}{B(0;T)}.$$

(b) By applying no arbitrage pricing principle again, the two portfolios constructed in (a) should have the same value at time t. We consider the following two cases:

- If $t < t_D$ and the t_D-year bond is not expired, then
$$f_t + KB(t;T) = S_t - DB(t;t_D) \Rightarrow f_t = S_t - DB(t;t_D) - KB(t;T).$$
- If $t \geq t_D$, the dividend received at t_D is used to repay the face value of the t_D-year bond and the portfolio 2 only contains 1 unit of the asset. So we have
$$f_t + KB(t;T) = S_t \Rightarrow f_t = S_t - KB(t;T).$$

Problem 18

When there is no arbitrage opportunity, the forward rate $f^T_{[0.5,0.75]}$ and the spot rates must satisfies

$$\underbrace{(1+0.05)^{0.5}}_{1+R_{[0,T_1]}}\underbrace{\left(1+f^T_{[0.5,0.75]}\right)^{0.25}}_{(1+K_{[T_1,T_2]})} = \underbrace{(1.06)^{0.75}}_{1+R_{[0,T_2]}} \Rightarrow f^T_{[0.5,0.75]} = 0.080287.$$

Since $f^T_{[0.5,0.75]} = 0.080287 > 0.07 = f_{[0.5,0.75]}$, there is arbitrage opportunity. We consider the following trading strategy:

- Borrow \$1 for 0.5 year and enter into FRA (as borrower) with forward rate $f_{[0.5,0.75]} = 0.07$. At time 0.5, refinance the repayment by borrowing $(1.05)^{0.5}$ for another 0.25 years.
- Invest \$1 received for 0.75 years.

The initial cost of the strategy is 0. On the other hand, the net cashflow at time 0.75 year is
$$Net\ cashflow = (1.06)^{0.75} - (1.05)^{0.5}(1.07)^{0.25} = 0.002496 > 0.$$
So the investor can get a positive profit for sure in future. So this is an arbitrage opportunity.

Problem 19

(a) When there is no arbitrage opportunity, the 6-month forward price should be given by
$$f_{[0,0.5]} = \frac{55}{B(0;0.5)} = 55(1.03)^{0.5} \approx 55.8189.$$
Since the market forward price (i.e. $56) is higher than no arbitrage forward price, one can construct an arbitrage opportunity as follows:

- Enter into 6-month forward contract (with delivery price \$56) in short position.
- Buy one unit of the underlying asset (at price $S_0 = \$55$)
- Short 6-month zero coupon bond with face value \$56.

The cost and the future payoff (after 6 months) of the strategy are summarized in the following table:

	Cost	Future payoff (after 6 months)
6-month forward contract in short position	0	$56 - S_T$
One unit of underlying asset	55	S_T

Short 6-month zero coupon bond	$-\frac{56}{(1.03)^{0.5}} = -55.17844$	-56
Net total	-0.17844	0

We observe that the investor receives an upfront payment of 0.17844 at time 0 and does not suffer from any potential loss in future. This is a desired arbitrage opportunity.

(b) When there is no arbitrage opportunity, the 6-month forward price should be given by

$$f_{[0,0.25]} = \frac{55}{B(0;0.5)} = 55(1.025)^{0.25} \approx 55.34057.$$

Since the market forward price (i.e. $\$56$) is higher than no arbitrage forward price, one can construct an arbitrage opportunity as follows:

- Enter into 3-month forward contract (with delivery price $56) in short position.
- Buy one unit of the underlying asset (at price $S_0 = \$55$)
- Short 3-month zero coupon bond with face value $56.

The cost and the future payoff (after 6 months) of the strategy are summarized in the following table:

	Cost	Future payoff (after 6 months)
6-month forward contract in short position	0	$56 - S_T$
One unit of underlying asset	55	S_T
Short 6-month zero coupon bond	$-\frac{56}{(1.025)^{0.25}} = -55.65537$	-56
Net total	-0.65537	0

We observe that the investor receives an upfront payment of 0.65537 at time 0 and does not suffer from any potential loss in future. This is a desired arbitrage opportunity.

Problem 20

To facilitate the discussion, we take Swiss Franc as foreign currency. When there is no arbitrage opportunity, the 3-month forward exchange rate (i.e. price of 1 unit of foreign currency in USD currency) must satisfy

$$f_{[0,0.25]} = \frac{S_0 B_F(0;0.25)}{B(0;0.25)} = \frac{1.0404\left(\frac{1}{e^{0.005(0.25)}}\right)}{\frac{1}{e^{0.01(0.25)}}} = 1.041701.$$

Since the market forward exchange rate (i.e. 1.03) is lower than $f_{[0,0.25]}$, one can consider the following arbitrage opportunity:

- Enter into 3-month forward contract in long position (i.e. exchange USD currency for Swiss Franc).
- Borrow $e^{-0.005(0.25)}$ unit of Swiss Franc and repay $e^{-0.005(0.25)}\left(e^{0.005(0.25)}\right) = 1$ unit of Swiss Franc after 3 months.

- Deposit $1.03e^{-0.01(0.25)}$ USD dollar.

The cost and the future payoff (after 6 months) of the strategy are summarized in the following table: (Here, S_T denotes the exchange rate)

	Cost	Future payoff (after 6 months)
3-month forward contract in long position	0	$S_T - 1.03$
Borrow $e^{-0.005(0.25)}$ unit of Swiss Franc	$-e^{-0.005(0.25)}(1.0404)$	$-S_T$
Deposit $1.03e^{-0.01(0.25)}$ USD dollar	$1.03e^{-0.01(0.25)}$	1.03
Net total	-0.01167	0

We observe that the investor receives an upfront payment of 0.01167 (in USD currency) at time 0 and does not suffer from any potential loss in future. This is a desired arbitrage opportunity.

Problem 21

(a) Recall that the forward price (delivery price) of the coupon bond is

$$f_{[0,T]} = \frac{S_0}{B(0;T)} - \frac{1}{B(0;T)} \sum_{i=1}^{k} rFB(0;iT_c) \dots \dots (*)$$

Note that

- ✓ 2 coupons (with amount $rF = 0.06(1000) = 60$) have been paid over the life of the forward contract (at time 1,2 respectively)
- ✓ Using the given yield rate, the current price of the bond is

$$S_0 = \sum_{k=1}^{10} \frac{1000(0.06)}{(1.05)^k} + \frac{1000}{(1.05)^{10}} \approx 1077.217.$$

From $(*)$, the delivery price can be computed as

$$f_{[0,2]} = \frac{1077.217}{\frac{1}{(1.049)^2}} - \frac{1}{\frac{1}{(1.049)^2}} \left(\frac{60}{1.045} + \frac{60}{(1.049)^2} \right) = 1062.19.$$

(b) We let f_1 be the contract value at time 1. Recall that

$$f_1 = S_1 - \underbrace{f_{[0,2]}}_{K} B(1;2) - 60B(1;2).$$

(*Note: Only 1 coupon is paid between time 1 and time 2)

Note that the bond price at time 1 is

$$S_1 = \sum_{k=2}^{10} \frac{60}{(1.048)^{k-1}} + \frac{1000}{(1.048)^9} = 1086.059.$$

Therefore, the value of the forward contract is

$$f_1 = 1086.059 - 1062.19 \left(\frac{1}{1.046} \right) - 60 \left(\frac{1}{1.046} \right) = 13.21961.$$

Problem 22

(a) We shall prove this statement using no arbitrage pricing principle. We consider the following two portfolios:

Portfolio 1: Enter into T-year forward contract in long position and buy a T-year zero coupon bond with face value K, where K is the delivery price.

(*Note: Then the portfolio 1 only contains 1 unit of the underlying asset at time T)

Portfolio 2: Buy 1 unit of the underlying asset). At the same time, we short-sell t_i-year zero coupon bond with face value D_i for $i = 1,2, \dots, n$. The dividend received from the asset at times $t_1, t_2, .., t_n$ are used to repay the face value of the bond. So the portfolio 2 only contains 1 unit of the underlying asset at time T.

(*Note: It is crucial to check the future payoffs of two portfolios since the use of no arbitrage pricing principle requires the fact that both portfolios have identical future payoff)

Since the future payoff of two portfolio are the same, it follows from no arbitrage pricing principle that the current value of these two portfolios must be the same, i.e.

$$f_0 + KB(0;T) = S_0 - \sum_{i=1}^{n} D_i B(0;t_i).$$

Since $f_0 = 0$ at the initialization date, we have

$$f_{[0,T]} = K = \frac{S_0}{B(0;T)} - \frac{1}{B(0;T)} \sum_{i=1}^{n} D_i B(0;t_i).$$

(b) Since no dividend is paid during the first 3 months, the 3-month forward price satisfies

$$\underbrace{42.8}_{f_{[0,0.25]}} = \underbrace{\frac{S_0}{\frac{1}{(1.06822672)^{0.25}}}}_{S_0/B(0;T)} \Rightarrow S_0 \approx 42.1.$$

Since two dividends are paid after 4 months and 8 months respectively, the 9-month forward price satisfies

$$\underbrace{42.003}_{f_{[0,0.75]}} = \underbrace{\frac{42.1}{\frac{1}{(1.06822672)^{0.75}}}}_{\frac{S_0}{B(0;T)}} - \underbrace{\frac{\frac{D}{(1.06822672)^{\frac{1}{3}}} + \frac{D}{(1.06822672)^{\frac{2}{3}}}}{\frac{1}{(1.06822672)^{0.75}}}}_{\frac{1}{B(0;T)}\sum_{i=1}^{n} D_i B(0;t_i)}$$

$\Rightarrow D = 1.09835.$

To calculate 6-month forward price, we note that there is only one dividend D paid over the first 6 months. Hence, we have

$$f_{[0,0.5]} = \frac{42.1}{\underbrace{\frac{1}{(1.06822672)^{0.5}}}_{S_0/B(0;T)}} - \frac{\frac{D}{(1.06822672)^{\frac{1}{3}}}}{\underbrace{\frac{1}{(1.06822672)^{0.5}}}_{\frac{1}{B(0;T)}\sum_{i=1}^{n} D_i B(0;t_i)}} \approx 42.40198.$$